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Decentralized Carbon Markets: Blockchain- Enabled Carbon Credit Exchange

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Introduction & Motivation

- Carbon credit trading is vital for climate change mitigation.
- Current systems are offline, inefficient, opaque, and centralized.
- Blockchain offers transparency, security, and decentralization.
- This research proposes a blockchain-based framework for carbon trading aligned with sustainability goals.

Scope of Study

- Focuses on enhancing carbon credit trading using blockchain technology.
- Emphasis on energy-efficient consensus mechanisms (PoS, DPoS, etc.).
- Integrates smart contracts, IoT, and layer-2 scalability solutions.
- Does not cover economic, political, or social factors.
- No implementation of a new blockchain network; theoretical and framework-based analysis.



Research Objective

Understand How Blockchain can change the Carbon Trading Markets for the better and Design a blockchain framework for efficient, transparent, and eco-friendly carbon credit trading.

Research Questions

1. How can blockchain enhance transparency and efficiency?
2. How can environmental challenges of blockchain be mitigated?
3. What consensus mechanisms are most suitable?
4. How can compliance and accessibility be ensured?
5. What role do smart contracts play?

Existing Carbon Markets

1. Compliance Markets: EU ETS, California Cap-and-Trade, China National ETS
 2. Voluntary Markets: VCS, Gold Standard, ACR, CAR
- Challenges: fraud, double counting, slow verification
 - Blockchain offers a unified, transparent platform



Limitations of Current System

1. Manual, error-prone verification
2. Lack of real-time transparency
3. High entry barriers for SMEs
4. Centralized intermediaries increase costs and reduce trust



Blockchain as a Solution

1. Decentralized ledger prevents tampering
2. Real-time traceability of credits
3. Removes need for intermediaries
4. Improves verification and accountability

Framework Components

User Authentication: Cryptography-based IDs

Tokenization: Digital carbon credit assets

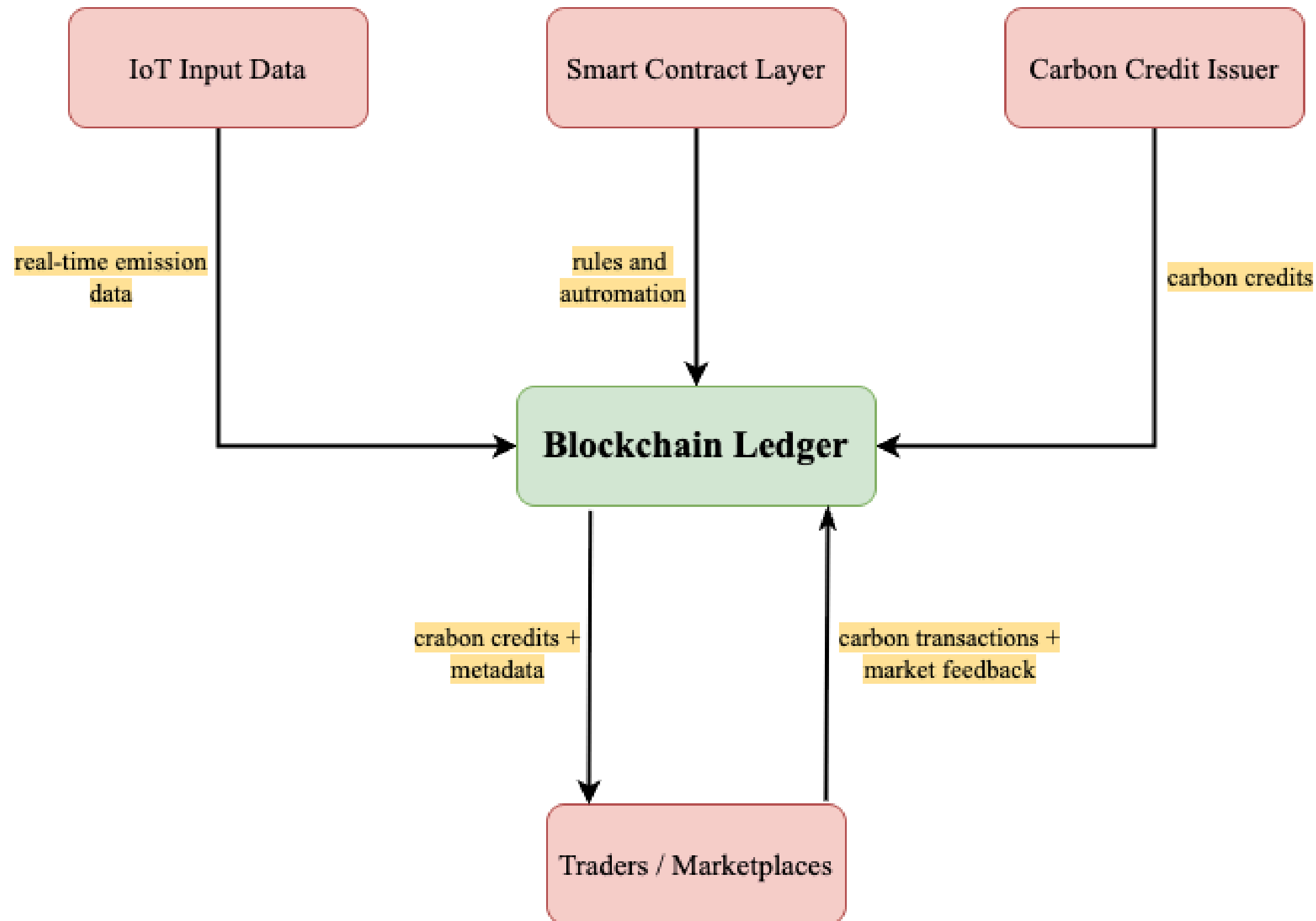
Smart Contracts: Automate trading and compliance

IoT Integration: Real-time data on emissions

Verification: Multi-party consensus

Interoperability: Cross-chain support

Flowchart of Proposed Framework

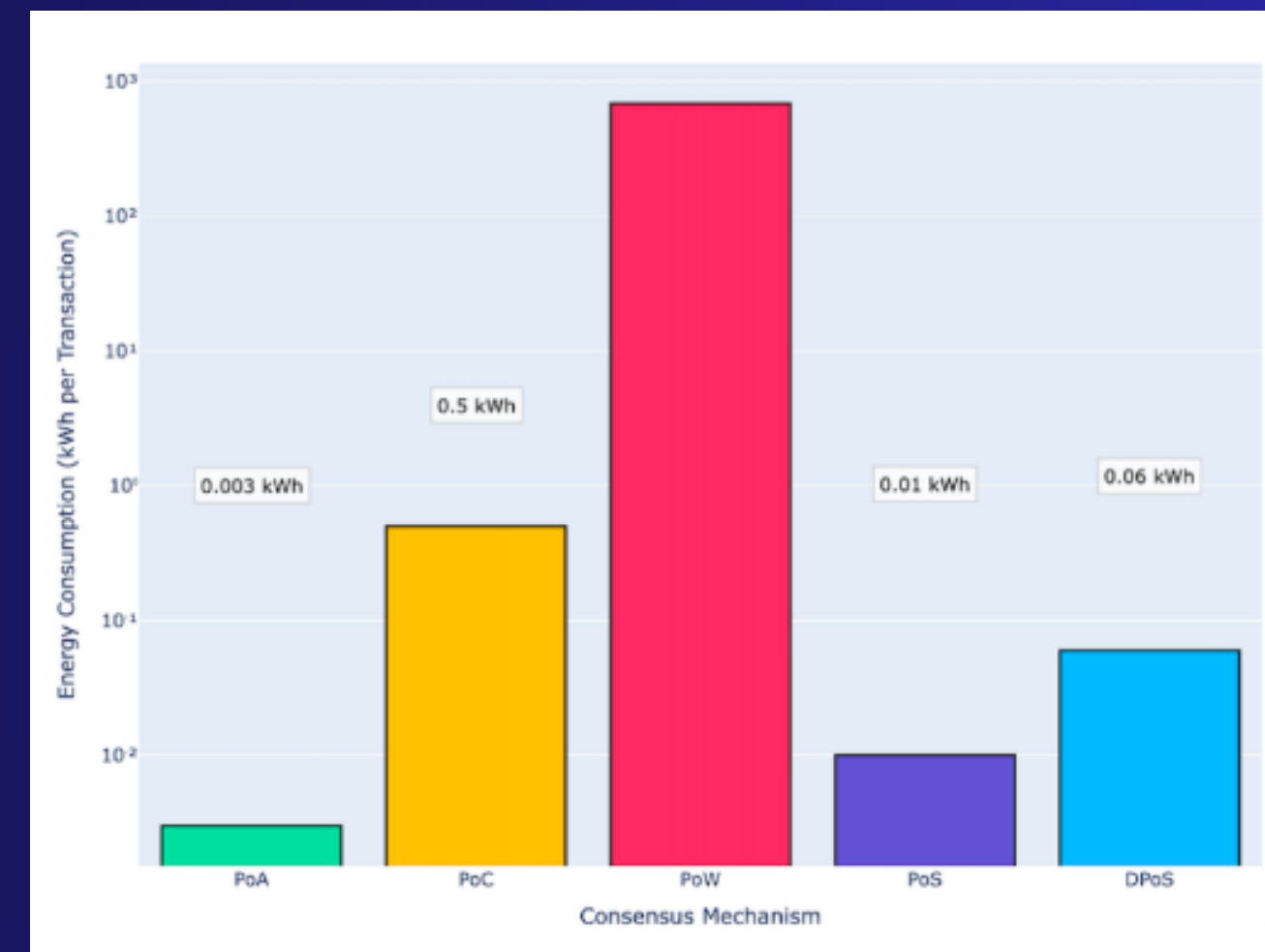


Smart Contracts in Action

- Auto-verification of emission reduction claims
- Fast, peer-to-peer trading
- Enforcement of compliance rules
- Reduced processing time and costs
- Transparent credit lifecycle tracking

Environmental Concerns

- PoW blockchains consume excessive energy
- Bitcoin: 215 KWh per transaction, 64.4 metric kilotons e-waste (2021)
- Irony: Energy-heavy tech to solve sustainability issues



Implementation Challenges

- Regulatory differences across countries
- Privacy vs. transparency balance
- Blockchain scalability and IoT integration
- Requires international standardization and frameworks

Future Vision & Impact

- Smart contracts + AI = intelligent trading
- Global inclusivity: SMEs & developing nations
- Increased trust, traceability, and liquidity
- Real-time market responsiveness



Conclusion

- Blockchain can revolutionize carbon markets
- Must balance innovation with eco-responsibility
- Smart contracts enhance efficiency and trust
- Requires collaboration between tech, regulators, and industry

Thank You

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